

Harkirat Singh

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EDUCATION

- **George Mason University** Fairfax, VA
Master of Science in Data Analytics Engineering Jan. 2024 – Present
- **University Institute of Engineering and Technology, Kurukshetra University** Kurukshetra, India
Bachelor of Technology in Electronics and Communication Engineering Aug. 2019 – June 2023

EXPERIENCE

- **George Mason University** Fairfax, VA
Research Analyst (RA) Jan 2025 – Present
 - **Data Analytics & Modeling:** Performed statistical and behavioral analysis on user engagement and cybersecurity learning trends using descriptive methods.
 - **Cybersecurity Research:** Led data workflows and analysis for a **12-week field study with 112 students**, measuring gamification impact on digital safety awareness.
 - **Behavioral & ML Insights:** Tracked **module completion (92.1%)**, session engagement (**24.2 mins**), and built ML models predicting phishing-detection success with **67.8%** accuracy.
 - **Manuscript Lead:** Co-led writing and statistical analysis for submitted ACM paper on gamified cyber-literacy intervention.
- **Falconet Internet Pvt. Ltd.** Kurukshetra, India
Data Analyst Dec 2022 – Dec 2023
 - **Database Migration:** Migrated datasets from Excel to PostgreSQL, improving data reliability by 20%.
 - **Dashboard Design:** Designed real-time Tableau dashboards for tracking KPIs.
 - **Risk Analysis:** Analyzed network performance, reducing downtime by 10%.
 - **Insights Reporting:** Generated reports using SQL and Tableau for better decision-making.

PROJECTS

- **GameSec: Gamified Cybersecurity Education — submitted to ACM CHI PLAY 2025:**
 - **Goal:** Designed a gamified curriculum to enhance cybersecurity awareness and engagement among PreK–12 students.
 - **Data Collection:** Collected pre/post quiz scores, gameplay logs, session times, and module completion data from **112 students** over 12 weeks.
 - **Analysis:** Calculated learning gains and behavioral patterns using descriptive statistics and aggregated metrics across age groups.
 - **Findings:** Observed improved quiz performance, increased engagement, and positive learning trends across all grade levels.
- **Bridge Longevity and Structural Integrity Research:**
 - **Objective:** Analyzed **13,331 bridges** built after 1970 to identify factors affecting deterioration, focusing on traffic volume, environmental exposure, and structural conditions.
 - **Methodology:** Applied Random Forest (**93% accuracy**) and Decision Tree (**90% accuracy**) models to predict bridge deterioration based on **deck condition, traffic volume, and environmental stress**.
 - **Findings:** Bridges **built before 1980** showed higher deterioration rates, and **proximity to water bodies** increased substructure damage risk.
 - **Outcome:** Developed predictive models to assist infrastructure agencies in prioritizing maintenance and optimizing bridge designs.

SKILLS

- **Languages:** Python, R, SQL, C++
- **Technologies:** AWS, Databricks, Hadoop, HDFS, MongoDB, PostgreSQL
- **Tools:** Tableau, Power BI, Microsoft Excel, PyCharm, Git
- **Methodologies:** Predictive Modeling, Risk Analysis, Statistical Analysis, ETL, Business Intelligence
- **Soft Skills:** Communication, Remote Collaboration, Presentation Skills, Critical Thinking, Project Leadership